

Supplementary materials for Rank-Poisson Transformation for use with Count Data in Poisson Regression

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4.3 Code for Simulation

The R code for the simulation is reported in this section. Others are encouraged to replicate and to adapt for other purposes.

```
k <- 10000; n <- 100
set.seed(033031)
rejects <- matrix(nrow=k,ncol=3*4+1)
rejects[,1] <- 1:k
group <- rep(0:1,each=n/2)
means2g <- matrix(nrow=k,ncol=6)
meanvarP <- matrix(nrow=k,ncol=12)
for (i in 1:k){
  ef0dpois <- c(makepois(n/2,lambda=4),makepois(n/2,lambda=4))
  datax <- data.frame(ef0dpois,group)
  means2g[i,1:2] <- c(diff(tapply(ef0dpois,group,mean)),
    sd_pooled(ef0dpois~group))
  meanvarP[i,1:2] <- tapply(ef0dpois,group,mean)
  meanvarP[i,3:4] <- tapply(ef0dpois,group,var)
  m0 <- glm(ef0dpois~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  rejects[i,2] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) rejects[i,2] <- FALSE

  m0 <- glm(poisrank(ef0dpois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  rejects[i,3] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) rejects[i,3] <- FALSE

  m0 <- glm(ef0dpois~1,family="quasipoisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  rejects[i,4] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) rejects[i,4] <- FALSE

  m0 <- lm(normrank(ef0dpois)~1,data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  rejects[i,5] <- comp$Pr[2] < .05
  if (comp$F[2] < .001) rejects[i,5] <- FALSE

  ef2dpois <- c(makepois(n/2,lambda=4),makepois(n/2,lambda=4.39))
  means2g[i,3:4] <- c(diff(tapply(ef2dpois,group,mean)),
    sd_pooled(ef2dpois~group))
  meanvarP[i,5:6] <- tapply(ef2dpois,group,mean)
  meanvarP[i,7:8] <- tapply(ef2dpois,group,var)

  datax <- data.frame(ef2dpois,group)
  m0 <- glm(ef2dpois~1,family="poisson")
  comp <- anova(m0,update(m0, .~. + group))
  rejects[i,6] <- comp$Pr[2] < .05
```

```

if (comp$Dev[2] < .001) rejects[i,6] <- FALSE

m0 <- glm(poisrank(ef2dpois)~1,family="poisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
rejects[i,7] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rejects[i,7] <- FALSE

m0 <- glm(ef2dpois~1,data=datax,family="quasipoisson")
comp <- anova(m0,update(m0, .~. + group))
rejects[i,8] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rejects[i,8] <- FALSE

m0 <- lm(normrank(ef2dpois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
rejects[i,9] <- comp$Pr[2] < .05
if (is.na(comp$F[2])) print(comp$F)
if (comp$F[2] < .001) rejects[i,9] <- FALSE

ef5dpois <- c(makepois(n/2,lambda=4),makepois(n/2,lambda=5.01))
means2g[i,5:6] <- c(diff(tapply(ef5dpois,group,mean)),
                    sd_pooled(ef5dpois~group))
meanvarP[i,9:10] <- tapply(ef5dpois,group,mean)
meanvarP[i,11:12] <- tapply(ef5dpois,group,var)

datax <- data.frame(ef5dpois,group)
m0 <- glm(ef5dpois~1,family="poisson")
comp <- anova(m0,update(m0, .~. + group))
rejects[i,10] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rejects[i,10] <- FALSE

m0 <- glm(poisrank(ef5dpois)~1,family="poisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
rejects[i,11] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rejects[i,11] <- FALSE

m0 <- glm(ef5dpois~1,family="quasipoisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
rejects[i,12] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rejects[i,12] <- FALSE

m0 <- lm(normrank(ef5dpois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
rejects[i,13] <- comp$Pr[2] < .05
if (comp$F[2] < .001) rejects[i,13] <- FALSE
}

```

```

k <- 10000; n <- 100
set.seed(03301)
means2exg <- matrix(nrow=k,ncol=6)
meanvarexP <- matrix(nrow=k,ncol=12)
exrejects <- matrix(nrow=k,ncol=3*4+1)
exrejects[,1] <- 1:k
group <- rep(0:1,each=n/2)

```

```

for (i in 1:k){
  ef0dexpois <- c(makeexpois(n/2,meanpois=4),
                  makeexpois(n/2,meanpois=4))
  datax <- data.frame(ef0dexpois,group)
  means2exg[i,1:2] <- c(diff(tapply(ef0dexpois,group,mean)),
                        sd_pooled(ef0dexpois~group))
  meanvarexP[i,1:2] <- tapply(ef0dexpois,group,mean)
  meanvarexP[i,3:4] <- tapply(ef0dexpois,group,var)
  m0 <- glm(ef0dexpois~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  exrejects[i,2] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) exrejects[i,2] <- FALSE

  m0 <- glm(poisrank(ef0dexpois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  exrejects[i,3] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) exrejects[i,3] <- FALSE

  m0 <- glm(ef0dexpois~1,family="quasipoisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  exrejects[i,4] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) exrejects[i,4] <- FALSE

  m0 <- lm(normrank(ef0dexpois)~1,data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  exrejects[i,5] <- comp$Pr[2] < .05
  if (comp$F[2] < .001) exrejects[i,5] <- FALSE

  ef2dexpois <- c(makeexpois(n/2,meanpois=4),
                  makeexpois(n/2,meanpois=4.69))
  means2exg[i,3:4] <- c(diff(tapply(ef2dexpois,group,mean)),
                        sd_pooled(ef2dexpois~group))
  meanvarexP[i,5:6] <- tapply(ef2dexpois,group,mean)
  meanvarexP[i,7:8] <- tapply(ef2dexpois,group,var)

  datax <- data.frame(ef2dexpois,group)
  m0 <- glm(ef2dexpois~1,family="poisson")
  comp <- anova(m0,update(m0, .~. + group))
  exrejects[i,6] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) exrejects[i,6] <- FALSE

  m0 <- glm(poisrank(ef2dexpois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  exrejects[i,7] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) exrejects[i,7] <- FALSE

  m0 <- glm(ef2dexpois~1,data=datax,family="quasipoisson")
  comp <- anova(m0,update(m0, .~. + group))
  exrejects[i,8] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) exrejects[i,8] <- FALSE

  m0 <- lm(normrank(ef2dexpois)~1,data=datax)
  comp <- anova(m0,update(m0, .~. + group))

```

```

exrejects[i,9] <- comp$Pr[2] < .05
if (comp$F[2] < .001) exrejects[i,9] <- FALSE

ef5dexpois <- c(makeexpois(n/2,meanpois=4),
               makeexpois(n/2,meanpois=5.73))
means2exg[i,5:6] <- c(diff(tapply(ef5dexpois,group,mean)),
                    sd_pooled(ef5dexpois~group))
meanvarxP[i,9:10] <- tapply(ef5dexpois,group,mean)
meanvarxP[i,11:12] <- tapply(ef5dexpois,group,var)

datax <- data.frame(ef5dexpois,group)
m0 <- glm(ef5dexpois~1,family="poisson")
comp <- anova(m0,update(m0, .~. + group))
exrejects[i,10] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) exrejects[i,10] <- FALSE

m0 <- glm(poisrank(ef5dexpois)~1,family="poisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
exrejects[i,11] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) exrejects[i,11] <- FALSE

m0 <- glm(ef5dexpois~1,family="quasipoisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
exrejects[i,12] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) exrejects[i,12] <- FALSE

m0 <- lm(normrank(ef5dexpois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
exrejects[i,13] <- comp$Pr[2] < .05
if (comp$F[2] < .001) exrejects[i,13] <- FALSE
}

```

```

k <- 10000; n <- 100
set.seed(565111)
means2rndg <- matrix(nrow=k,ncol=6)
meanvarrndP <- matrix(nrow=k,ncol=12)
rndrejects <- matrix(nrow=k,ncol=3*4+1)
rndrejects[,1] <- 1:k
group <- rep(0:1,each=n/2)
for (i in 1:k){
  ef0drndpois <- c(roundish(makepois(n/2,lambda=4)),
                  roundish(makepois(n/2,lambda=4)))
  means2rndg[i,1:2] <- c(diff(tapply(ef0drndpois,group,mean)),
                        sd_pooled(ef0drndpois~group))
  meanvarrndP[i,1:2] <- tapply(ef0drndpois,group,mean)
  meanvarrndP[i,3:4] <- tapply(ef0drndpois,group,var)
  datax <- data.frame(ef0drndpois,group)
  m0 <- glm(ef0drndpois~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  rndrejects[i,2] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) rndrejects[i,2] <- FALSE

  m0 <- glm(poisrank(ef0drndpois)~1,family="poisson",data=datax)

```

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comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,3] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,3] <- FALSE

m0 <- glm(ef0drndpois~1,family="quasipoisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,4] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,4] <- FALSE

m0 <- lm(normrank(ef0drndpois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,5] <- comp$Pr[2] < .05
if (comp$F[2] < .001) rndrejects[i,5] <- FALSE

ef2drndpois <- c(roundish(makepois(n/2,lambda=4)),
                 roundish(makepois(n/2,lambda=5.45)))
means2rndg[i,3:4] <- c(diff(tapply(ef2drndpois,group,mean)),
                      sd_pooled(ef2drndpois~group))
meanvarrndP[i,5:6] <- tapply(ef2drndpois,group,mean)
meanvarrndP[i,7:8] <- tapply(ef2drndpois,group,var)
datax <- data.frame(ef2drndpois,group)
m0 <- glm(ef2drndpois~1,family="poisson")
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,6] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,6] <- FALSE

m0 <- glm(poisrank(ef2drndpois)~1,family="poisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,7] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,7] <- FALSE

m0 <- glm(ef2drndpois~1,data=datax,family="quasipoisson")
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,8] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,8] <- FALSE

m0 <- lm(normrank(ef2drndpois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,9] <- comp$Pr[2] < .05
if (comp$F[2] < .001) rndrejects[i,9] <- FALSE

ef5drndpois <- c(roundish(makepois(n/2,lambda=4)),
                 roundish(makepois(n/2,lambda=7.69)))
means2rndg[i,5:6] <- c(diff(tapply(ef5drndpois,group,mean)),
                      sd_pooled(ef5drndpois~group))
meanvarrndP[i,9:10] <- tapply(ef5drndpois,group,mean)
meanvarrndP[i,11:12] <- tapply(ef5drndpois,group,var)

datax <- data.frame(ef5drndpois,group)
m0 <- glm(ef5drndpois~1,family="poisson")
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,10] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,10] <- FALSE

```

```

m0 <- glm(poisrank(ef5drndpois)~1,family="poisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,11] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,11] <- FALSE

m0 <- glm(ef5drndpois~1,family="quasipoisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,12] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) rndrejects[i,12] <- FALSE

m0 <- lm(normrank(ef5drndpois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
rndrejects[i,13] <- comp$Pr[2] < .05
if (comp$F[2] < .001) rndrejects[i,13] <- FALSE
}

```

```

k <- 10000; n <- 100
set.seed(5929)
brrejects <- matrix(nrow=k,ncol=3*4+1)
meanvarbrP <- matrix(nrow=k,ncol=12)
brrejects[,1] <- 1:k
means2brg <- matrix(nrow=k,ncol=6)
group <- rep(0:1,each=n/2)
for (i in 1:k){
  ef0dbrpois <- c(br(makepois(n/2,lambda=4)),
                  br(makepois(n/2,lambda=4)))
  means2brg[i,1:2] <- c(diff(tapply(ef0dbrpois,group,mean)),
                        sd_pooled(ef0dbrpois~group))
  meanvarbrP[i,1:2] <- tapply(ef0dbrpois,group,mean)
  meanvarbrP[i,3:4] <- tapply(ef0dbrpois,group,var)
  datax <- data.frame(ef0dbrpois,group)
  m0 <- glm(ef0dbrpois~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,2] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,2] <- FALSE

  m0 <- glm(poisrank(ef0dbrpois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,3] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,3] <- FALSE

  m0 <- glm(ef0dbrpois~1,family="quasipoisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,4] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,4] <- FALSE

  m0 <- lm(normrank(ef0dbrpois)~1,data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,5] <- comp$Pr[2] < .05
  if (comp$F[2] < .001) brrejects[i,5] <- FALSE

  ef2dbrpois <- c(br(makepois(n/2,lambda=4)),

```

```

      br(makepois(n/2,lambda=4.40)))
means2brg[i,3:4] <- c(diff(tapply(ef2dbrpois,group,mean)),
  sd_pooled(ef2dbrpois~group))
meanvarbrP[i,5:6] <- tapply(ef2dbrpois,group,mean)
meanvarbrP[i,7:8] <- tapply(ef2dbrpois,group,var)
datax <- data.frame(ef2dbrpois,group)
  m0 <- glm(ef2dbrpois~1,family="poisson")
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,6] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,6] <- FALSE

  m0 <- glm(poisrank(ef2dbrpois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,7] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,7] <- FALSE

  m0 <- glm(ef2dbrpois~1,data=datax,family="quasipoisson")
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,8] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,8] <- FALSE

  m0 <- lm(normrank(ef2dbrpois)~1,data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,9] <- comp$Pr[2] < .05
  if (comp$F[2] < .001) brrejects[i,9] <- FALSE

ef5dbrpois <- c(br(makepois(n/2,lambda=4)),
  br(makepois(n/2,lambda=5.01)))
  means2brg[i,5:6] <- c(diff(tapply(ef5dbrpois,group,mean)),
    sd_pooled(ef5dbrpois~group))
meanvarbrP[i,9:10] <- tapply(ef5dbrpois,group,mean)
meanvarbrP[i,11:12] <- tapply(ef5dbrpois,group,var)

  datax <- data.frame(ef5dbrpois,group)
  m0 <- glm(ef5dbrpois~1,family="poisson")
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,10] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,10] <- FALSE

  m0 <- glm(poisrank(ef5dbrpois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,11] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,11] <- FALSE

  m0 <- glm(ef5dbrpois~1,family="quasipoisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,12] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) brrejects[i,12] <- FALSE

  m0 <- lm(normrank(ef5dbrpois)~1,data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  brrejects[i,13] <- comp$Pr[2] < .05
  if (comp$F[2] < .001) brrejects[i,13] <- FALSE

```

```
}
```

```
k <- 10000; n <- 100
set.seed(592219)
means2zig <- matrix(nrow=k,ncol=6)
meanvarziP <- matrix(nrow=k,ncol=12)
zirejects <- matrix(nrow=k,ncol=3*4+1)
zirejects[,1] <- 1:k
group <- rep(0:1,each=n/2)
for (i in 1:k){
  ef0dzipois <- c(zi(makepois(n/2,lambda=4)),
                 zi(makepois(n/2,lambda=4)))
  means2zig[i,1:2] <- c(diff(tapply(ef0dzipois,group,mean)),
                       sd_pooled(ef0dzipois~group))
  meanvarziP[i,1:2] <- tapply(ef0dzipois,group,mean)
  meanvarziP[i,3:4] <- tapply(ef0dzipois,group,var)

  datax <- data.frame(ef0dzipois,group)
  m0 <- glm(ef0dzipois~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  zirejects[i,2] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) zirejects[i,2] <- FALSE

  m0 <- glm(poisrank(ef0dzipois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  zirejects[i,3] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) zirejects[i,3] <- FALSE

  m0 <- glm(ef0dzipois~1,family="quasipoisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  zirejects[i,4] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) zirejects[i,4] <- FALSE

  m0 <- lm(normrank(ef0dzipois)~1,data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  zirejects[i,5] <- comp$Pr[2] < .05
  if (comp$F[2] < .001) zirejects[i,5] <- FALSE

  ef2dzipois <- c(zi(makepois(n/2,lambda=4)),
                 zi(makepois(n/2,lambda=4.85)))
  means2zig[i,3:4] <- c(diff(tapply(ef2dzipois,group,mean)),
                       sd_pooled(ef2dzipois~group))
  meanvarziP[i,5:6] <- tapply(ef2dzipois,group,mean)
  meanvarziP[i,7:8] <- tapply(ef2dzipois,group,var)
  datax <- data.frame(ef2dzipois,group)
  m0 <- glm(ef2dzipois~1,family="poisson")
  comp <- anova(m0,update(m0, .~. + group))
  zirejects[i,6] <- comp$Pr[2] < .05
  if (comp$Dev[2] < .001) zirejects[i,6] <- FALSE

  m0 <- glm(poisrank(ef2dzipois)~1,family="poisson",data=datax)
  comp <- anova(m0,update(m0, .~. + group))
  zirejects[i,7] <- comp$Pr[2] < .05
```

```

if (comp$Dev[2] < .001) zirejects[i,7] <- FALSE

m0 <- glm(ef2dzipois~1,data=datax,family="quasipoisson")
comp <- anova(m0,update(m0, .~. + group))
zirejects[i,8] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) zirejects[i,8] <- FALSE

m0 <- lm(normrank(ef2dzipois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
zirejects[i,9] <- comp$Pr[2] < .05
if (comp$F[2] < .001) zirejects[i,9] <- FALSE

ef5dzipois <- c(zi(makepois(n/2,lambda=4)),
               zi(makepois(n/2,lambda=6.08)))
means2zig[i,5:6] <- c(diff(tapply(ef5dzipois,group,mean)),
                      sd_pooled(ef5dzipois~group))
meanvarziP[i,9:10] <- tapply(ef5dzipois,group,mean)
meanvarziP[i,11:12] <- tapply(ef5dzipois,group,var)

datax <- data.frame(ef5dzipois,group)
m0 <- glm(ef5dzipois~1,family="poisson")
comp <- anova(m0,update(m0, .~. + group))
zirejects[i,10] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) zirejects[i,10] <- FALSE

m0 <- glm(poisrank(ef5dzipois)~1,family="poisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
zirejects[i,11] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) zirejects[i,11] <- FALSE

m0 <- glm(ef5dzipois~1,family="quasipoisson",data=datax)
comp <- anova(m0,update(m0, .~. + group))
zirejects[i,12] <- comp$Pr[2] < .05
if (comp$Dev[2] < .001) zirejects[i,12] <- FALSE

m0 <- lm(normrank(ef5dzipois)~1,data=datax)
comp <- anova(m0,update(m0, .~. + group))
zirejects[i,13] <- comp$Pr[2] < .05
if (comp$F[2] < .001) zirejects[i,13] <- FALSE
}

```

4.4 Choosing Defaults for poisrank function

There are two options in the `poisrank` function. Simulations are reported in this section to explore what default values should be used and the impact of different choices. The first option, labelled `delta`, is also included in the `normrank` function.[18] The values here should be less than one (i.e., if $\delta = 1$ a single maximum will result in applying `qpois(1,x)` which produces ∞ , which the R `as.integer` function coerces to `NA`). The first simulation in this section reports how the choice of δ affects how well the transformation produces a variable that approximates the Poisson distribution.

The second option relates to the mean used for the R quantile Poisson function (`qpois`), called `lambda` in that function. The mean of the observed data seems a good choice for the default, though a few high observed values—as often occurs with overdispersed data—can make the mean very high, and this can lead to computational issues (issues also with λ values near zero). The `poisrank` function created here allows the

mean to be trimmed, with the default set to no trimming.

Data were created from a Poisson distribution with $\lambda = 4$ and from a negative binomial distribution with `rnbinom` size of 6 and probability of .5, which produces a mean of 6 and variance of 12. Further, zero-inflated data, data with response prototypes (more responses at multiples of five), and data with fewer responses in some regions (i.e., gaps) are created from the Poisson data. These distributions are meant to represent some of the ways in which count variables can be distributed. The goal is to see if there is some consistency across these. If so, further distributions should be examined. The rounded distribution, the distribution with gaps, and the zero-inflated distribution are created with the following functions using the defaults.

```
lambda1 <- 1.5 #
lambda2 <- .3  # 0 it is Poisson
numb <- 100
means <- vars <- vector(length=numb)
for (i in 1:numb){
  x <- rgenpois(1000, lambda1, lambda2)
  means[i] <- mean(x)
  vars[i] <- var(x) }

makeexpois <- function(n,varpois=2,meanpois=4,cov1=FALSE,cov2=FALSE,
                      cov3=FALSE,cov4=FALSE){
  size <- varpois
  ifelse (cov1, cov1 <- runif(n,-meanpois,meanpois), cov1 <- 0)
  ifelse (cov2, cov2 <- runif(n,-meanpois,meanpois), cov2 <- 0)
  ifelse (cov3, cov3 <- runif(n,.25,4), cov3 <- 1)
  ifelse (cov4, cov4 <- runif(n,.25,4), cov4 <- 1)
  mu <- (cov3*cov4*meanpois+cov1+cov2) #+ 2*rnbinom(n,1,.5) # a little extra error
  prob <- size/(size+mu)
  rnbinom(n,size=size,prob=prob)
}

makepois <- function(n,lambda=4,cov1=FALSE,cov2=FALSE,
                    cov3=FALSE,cov4=FALSE)
{ ifelse (cov1, cov1 <- runif(n,-lambda,lambda), cov1 <- 0)
  ifelse (cov2, cov2 <- runif(n,-lambda,lambda), cov2 <- 0)
  ifelse (cov3, cov3 <- runif(n,.25,4), cov3 <- 1)
  ifelse (cov4, cov4 <- runif(n,.25,4), cov4 <- 1)
  lambda <- (cov3*cov4*lambda+cov1+cov2) #+ 2*rnbinom(n,1,.5) # a little extra error
  rpois(n,lambda)
}
```

```
x1a <- makeexpois(1000,varpois=1000000)
x2a <- makeexpois(1000,varpois=2)
x3a <- makeexpois(1000,varpois=1000000,cov1=TRUE)
x4a <- makeexpois(1000,varpois=1000000,cov3=TRUE)

x1b <- makepois(1000)
x2b <- makepois(1000,cov1=TRUE)
x3b <- makepois(1000,cov3=TRUE)
```

With rank-based functions the rank is often divided by the the sample size (n) to create percentiles. Often a small $\delta > 0$ is added to the ranks to prevent a single maximum having the value 1, which for some functions, including `qnorm` and `qpois`, this can be problematic (it results in ∞). Several non-normal distributions $\delta = .7$ worked fairly well when transforming the ranks to approximate a normal distribution,

but other values between 0 and 1 have also be argued for.[18] The default for the `blom` function from **rcompanion**[10] is $\pi/8 \approx .39$. Here, it is examined if there is some value that works well across several non-Poisson distributions. The simulation will start at $\delta = .001$ and test values up to $\delta = .999$. The sample size was set to $n = 100$.

```
n <- 100
set.seed(2222)
x <- rpois(n,4)
expx <- rnbinom(n,6,.5)
brx <- br(x)
rx <- roundish(x)
zix <- zi(x)
prx <- poisrank(x)
prexpx <- poisrank(expx)
prbrx <- poisrank(brx)
prrx <- poisrank(rx)
przix <- poisrank(zix)
cors <- diag(cor(cbind(x,expx,brx,rx,zix),cbind(prx,prexpx,prbrx,prrx,przix)))
```

```
n <- 100
k <- 1001
set.seed(3728)
dds <- seq(0.001,.999,.001)
vals <- matrix(ncol=12,nrow=k*length(dds))
set.seed(432)
rownum <- 0
for (i in 1:k)
  for (dd in 1:length(dds)) {
    rownum <- rownum + 1
    vals[rownum,1] <- rownum
    vals[rownum,2] <- dds[dd]
    x <- rpois(n,3)
    expx <- makeexpois(n,meanpois=3)
    brx <- br(x)
    rx <- roundish(x)
    zix <- zi(x)
    vals[rownum,3] <- poisson.m(x)[2]
    vals[rownum,4] <- poisson.m(poisrank(x,delta=dds[dd]))[2]
    vals[rownum,5] <- poisson.m(expx)[2]
    vals[rownum,6] <- poisson.m(poisrank(expx,delta=dds[dd]))[2]
    brx <- as.integer(brx)
    vals[rownum,7] <- poisson.m(brx)[2]
    vals[rownum,8] <- poisson.m(poisrank(brx,delta=dds[dd]))[2]
    vals[rownum,9] <- poisson.m(rx)[2]
    vals[rownum,10] <- poisson.m(poisrank(rx,delta=dds[dd]))[2]
    vals[rownum,11] <- poisson.m(zix)[2]
    vals[rownum,12] <- poisson.m(poisrank(zix,delta=dds[dd]))[2]
  }
colnames(vals) <- c("trial","delta","pois","PRpois","expois",
                  "PRexpois","brx","PRbrx","rx","PRrx",
                  "zix","PRzix")
```

To estimate how well the resulting distribution resembles the Poisson distribution the mean-deviation test,[12] as implemented in the **energy** package,[11] is used. The Anderson-Darling distance metric is used.[1]

Table 4 shows that the Poisson-rank transformation leads to the variable better approximating the Poisson distribution, except for when the original data are drawn from a Poisson distribution.

```

trx <- 0
prpois <- tapply(vals[,4],vals[,2],mean,tr=trx,na.rm=TRUE)
prexppois <- tapply(vals[,6],vals[,2],mean,tr=trx,na.rm=TRUE)
prbrx <- tapply(vals[,8],vals[,2],mean,tr=trx,na.rm=TRUE)
prrx <- tapply(vals[,10],vals[,2],mean,tr=trx,na.rm=TRUE)
przix <- tapply(vals[,12],vals[,2],mean,tr=trx,na.rm=TRUE)
prpoisM <- min(tapply(vals[,4],vals[,2],mean),na.rm=TRUE)
dpoisM <- vals[which.min(tapply(vals[,4],vals[,2],mean,na.rm=TRUE)),2]
prexppoisM <- min(tapply(vals[,6],vals[,2],mean),na.rm=TRUE)
dexppoisM <- vals[which.min(tapply(vals[,6],vals[,2],mean,na.rm=TRUE)),2]
prbrxM <- min(tapply(vals[,8],vals[,2],mean),na.rm=TRUE)
dbrxM <- vals[which.min(tapply(vals[,8],vals[,2],mean,na.rm=TRUE)),2]
prrxM <- min(tapply(vals[,10],vals[,2],mean),na.rm=TRUE)
drxM <- vals[which.min(tapply(vals[,10],vals[,2],mean,na.rm=TRUE)),2]
przixM <- min(tapply(vals[,12],vals[,2],mean),na.rm=TRUE)
dzixM <- vals[which.min(tapply(vals[,12],vals[,2],mean,na.rm=TRUE)),2]

# The following aren't affected by delta. Calculated
# here in case anyone wants to compare the same distributions
# with and without the transformation.
pois <- tapply(vals[,3],vals[,2],mean,tr=trx,na.rm=TRUE)
exppois <- tapply(vals[,5],vals[,2],mean,tr=trx,na.rm=TRUE)
brx <- tapply(vals[,7],vals[,2],mean,tr=trx,na.rm=TRUE)
rx <- tapply(vals[,9],vals[,2],mean,tr=trx,na.rm=TRUE)
zix <- tapply(vals[,11],vals[,2],mean,tr=trx,na.rm=TRUE)

print(xtable(xxm,caption="The mean Anderson-Darling distance
  between the distribution and the Poisson
  distribution for different  $\delta$  values.
  Low values mean the distribution is closer to the Poisson distribution.
  The value for untransformed Poisson is what you would expect if
  the transformation made each of these distributions like they were
  drawn from a Poisson distribution. \\vspace{.2cm}",
  digits=3,label="tab:xxm"),caption.placement="top")

```

Table 4: The mean Anderson-Darling distance between the distribution and the Poisson distribution for different δ values. Low values mean the distribution is closer to the Poisson distribution. The value for untransformed Poisson is what you would expect if the transformation made each of these distributions like they were drawn from a Poisson distribution.

	pois	exppois	gaps	rounding	zeroInf
raw	1.112	42.700	1.117	51.648	39.686
transformed	0.902	4.820	0.905	1.297	10.204
min dist	0.786	3.708	0.788	0.848	8.820
lowest delta	0.731	0.995	0.731	0.999	0.999

Once the Poisson rank transformation is applied the distances drop for all but those data drawn from a Poisson distribution. Figure 1 shows the trimmed means of the Anderson-Darling distance for each δ of these transformed distributions from the Poisson distribution. The curves have been smoothed using the

loess function with a span of .1.[5] The relationship between δ and the distance from Poisson appears to decrease, but only slightly. The default is set near zero (`delta = .0001`).

```
par(mar=c(4,7,2,2))
spanx <- .1
l2prpois <- predict(loess(prpois~dds,span=spanx))
l2prexppois <- predict(loess(prexppois~dds,span=spanx))
l2prbrx <- predict(loess(prbrx~dds,span=spanx))
l2prrx <- predict(loess(prrx~dds,span=spanx))
l2przix <- predict(loess(przix~dds,span=spanx),newdata=data.frame(dds=dds))
#length(l2przix)
#length(dds)

plot(dds,l2prpois,ylim=c(0,16),xlim=c(0,1.03),col="white",las=1,
      ylab="",xlab=expression(delta),xaxt='n')
axis(1,seq(0,1,.2),c("0.0","0.2","0.4","0.6","0.8","1.0"))
mtext("Trimmed Mean of Distances from Poisson",2,3.7)
mtext("Anderson-Darling metric",2,2.5)
lines(dds,l2prpois,col=2,lwd=2.5)
lines(dds,l2prexppois,col=3,lwd=2.5)
lines(dds,l2prbrx,col=4,lwd=2.5)
lines(dds,l2prrx,col=5,lwd=2.5)
lines(dds,l2przix,col=6,lwd=2.5)
dists <- c("Zero-Inf Poisson","Poisson w/ rounding",
           "Extra-Poisson","Poisson w/ Gaps","Poisson")
labs <- as.factor(dists)
levels(labs) <- dists
legend("topright",legend=labs,col=c(6,5,3,4,2),lwd=3,cex=.9)
```

Data were created in a similar manner to explore the effect of different trims of the observed mean for the λ used to create the transformed data. The `qpois` function requires the mean for the Poisson distribution to be entered. The mean of the observed variable seems a good option. However, means can be greatly affected by extreme values and these often occur with overdispersed count variables.

Trimmed means are less affected by extreme values than traditional untrimmed means. Galton, for example, argues for the median (the 50% trimmed mean) because this corresponds with half the values being above the estimate and half below.[7] Wilcox argues that a 20% trimmed mean provides a good level of trimming for many circumstances.[17] We did not know if the amount of trimming would make a large difference for the `poisrank` function when it estimates the Poisson mean. We examine values of it from .0 (the mean) to .5 (the median). Tables ?? and ?? show minimal effect of the choice for the trim, so the default is set to 0 (see also Figure 2).

```
n <- 100
k <- 1001
set.seed(29405)
dds <- seq(0,.5,.01)
vals <- matrix(ncol=12,nrow=k*length(dds))
set.seed(332)
rownum <- 0
for (i in 1:k){
  for (dd in 1:length(dds)){
    rownum <- rownum + 1
    vals[rownum,1] <- rownum
    vals[rownum,2] <- dds[dd]
    x <- rpois(n,3)
```

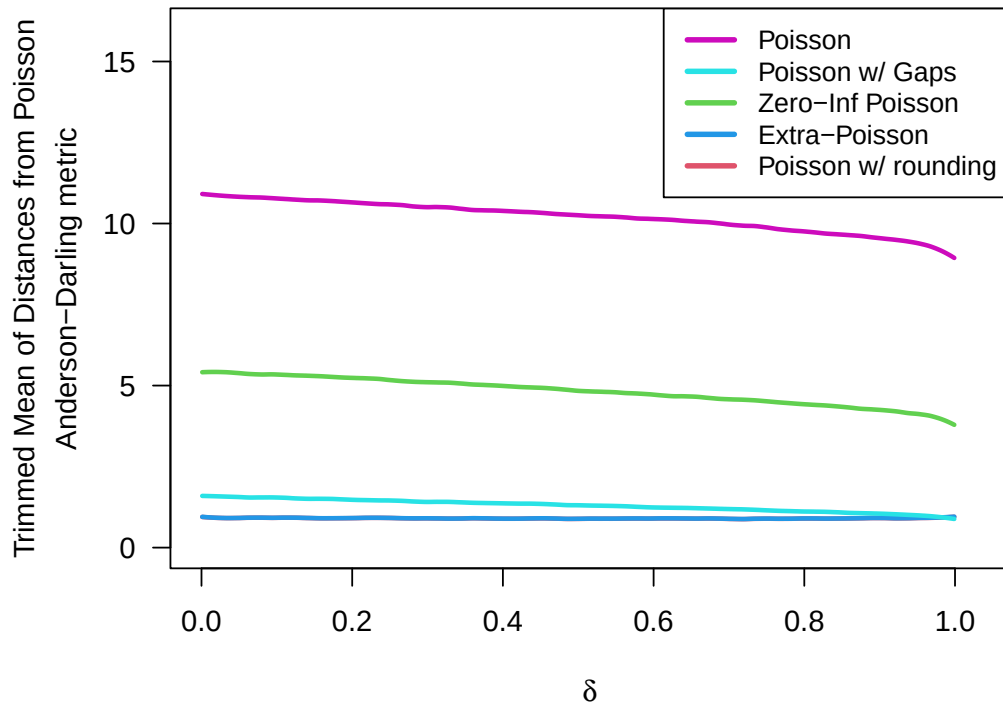


Figure 1: The difference between the Poisson distribution and the different distributions tested, after the Poisson rank transformation has been applied, for different δ values. The Anderson-Darling metric is used. The data are smoothed using the `loess` with a span of .1.

```

expx <- makeexpois(n,meanpois=3)
brx <- br(x)
brx <- as.integer(brx)
rx <- roundish(x)
zix <- zi(x)
vals[rownum,3] <- poisson.m(x)[2]
vals[rownum,4] <- poisson.m(poisrank(x,tr=dd))[2]
vals[rownum,5] <- poisson.m(expx)[2]
vals[rownum,6] <- poisson.m(poisrank(expx,tr=dd))[2]
vals[rownum,7] <- poisson.m(brx)[2]
vals[rownum,8] <- poisson.m(poisrank(brx,tr=dd))[2]
vals[rownum,9] <- poisson.m(rx)[2]
vals[rownum,10] <- poisson.m(poisrank(rx,tr=dd))[2]
vals[rownum,11] <- poisson.m(zix)[2]
vals[rownum,12] <- poisson.m(poisrank(zix,tr=dd))[2]
}
colnames(vals) <- c("trial","delta","pois","PRpois","expois",
  "PRexpois","brx","PRbrx","rx","PRrx",
  "zix","PRzix")

```

```

prpois <- tapply(vals[,4],vals[,2],mean,tr=0)
prexppois <- tapply(vals[,6],vals[,2],mean,tr=0)
prbrx <- tapply(vals[,8],vals[,2],mean,tr=0)
prrx <- tapply(vals[,10],vals[,2],mean,tr=0)
przix <- tapply(vals[,12],vals[,2],mean,tr=0)
prpoisM <- min(tapply(vals[,4],vals[,2],mean),na.rm=TRUE)
dpoisM <- vals[which.min(tapply(vals[,4],vals[,2],mean,na.rm=TRUE)),2]
prexppoisM <- min(tapply(vals[,6],vals[,2],mean),na.rm=TRUE)
dexppoisM <- vals[which.min(tapply(vals[,6],vals[,2],mean,na.rm=TRUE)),2]
prbrxM <- min(tapply(vals[,8],vals[,2],mean),na.rm=TRUE)
dbrxM <- vals[which.min(tapply(vals[,8],vals[,2],mean,na.rm=TRUE)),2]
prrxM <- min(tapply(vals[,10],vals[,2],mean),na.rm=TRUE)
drxM <- vals[which.min(tapply(vals[,10],vals[,2],mean,na.rm=TRUE)),2]
przixM <- min(tapply(vals[,12],vals[,2],mean),na.rm=TRUE)
dzixM <- vals[which.min(tapply(vals[,12],vals[,2],mean,na.rm=TRUE)),2]
# The following aren't affected by delta, but calculated
# here in case anyone wants to compare the same distributions
# with and without the transformation
pois <- tapply(vals[,3],vals[,2],mean,tr=0)
exppois <- tapply(vals[,5],vals[,2],mean,tr=0)
brx <- tapply(vals[,7],vals[,2],mean,tr=0)
rx <- tapply(vals[,9],vals[,2],mean,tr=0)
zix <- tapply(vals[,11],vals[,2],mean,tr=0)

```

```

xxm <- matrix(apply(vals[,2:6],nrow=2),nrow=2)[,2:6]
xxm <- rbind(xxm,
  c(prpoisM,prexppoisM,prbrxM,prrxM,przixM),
  c(dpoisM,dexppoisM,dbrxM,drxM,dzixM))
xxtm <- matrix(apply(vals[,2:6],nrow=2),nrow=2)[,2:6]
xxtm <- rbind(xxtm,c(
  min(prpois,na.rm=TRUE),min(prexppois,na.rm=TRUE),
  min(prbrx,na.rm=TRUE),min(prrx,na.rm=TRUE),min(przix,na.rm=TRUE)),

```

```

c(vals[which.min(prpois),2],
  vals[which.min(prexpoids),2],
  vals[which.min(prbrx),2],
  vals[which.min(prrx),2],
  vals[which.min(przix),2]))
rownames(xxm) <- rownames(xxtm) <- c("raw","transformed",
                                     "min dist","lowest delta")
colnames(xxm) <- colnames(xxtm) <-
  c("pois","expoids","gaps","rounding","zeroInf")

```

```

par(mar=c(6,7,2,2))
spanx <- .2
l2prpois <- predict(loess(prpois~dds,span=spanx))
l2prexpoids <- predict(loess(prexpoids~dds,span=spanx))
l2prbrx <- predict(loess(prbrx~dds,span=spanx))
l2prrx <- predict(loess(prrx~dds,span=spanx))
l2przix <- predict(loess(przix~dds,span=spanx),newdata=data.frame(dds=dds))
#length(l2przix)

plot(dds,l2prpois,ylim=c(0,15),col="white",las=1,
      ylab="",xlab="Amount of trim")
mtext("Trimmed Mean of Distances from Poisson",2,4.2)
mtext("Anderson-Darling metric",2,3)
lines(dds,l2prpois,col=2)
lines(dds,l2prexpoids,col=3)
lines(dds,l2prbrx,col=4)
lines(dds,l2prrx,col=5)
lines(dds,l2przix,col=6)
leg_ord <- matrix(1:6,nrow=2,byrow=TRUE)
legend("top",c("Pois","Expoids","Gap","round","ZeroInf","")[leg_ord],
      col=c(2:6,0)[leg_ord],lwd=rep(2.5,6)[leg_ord],ncol=3)
mtext("Mean",1,2.7,at=0)
mtext("Median",1,2.7,at=0.5)

```

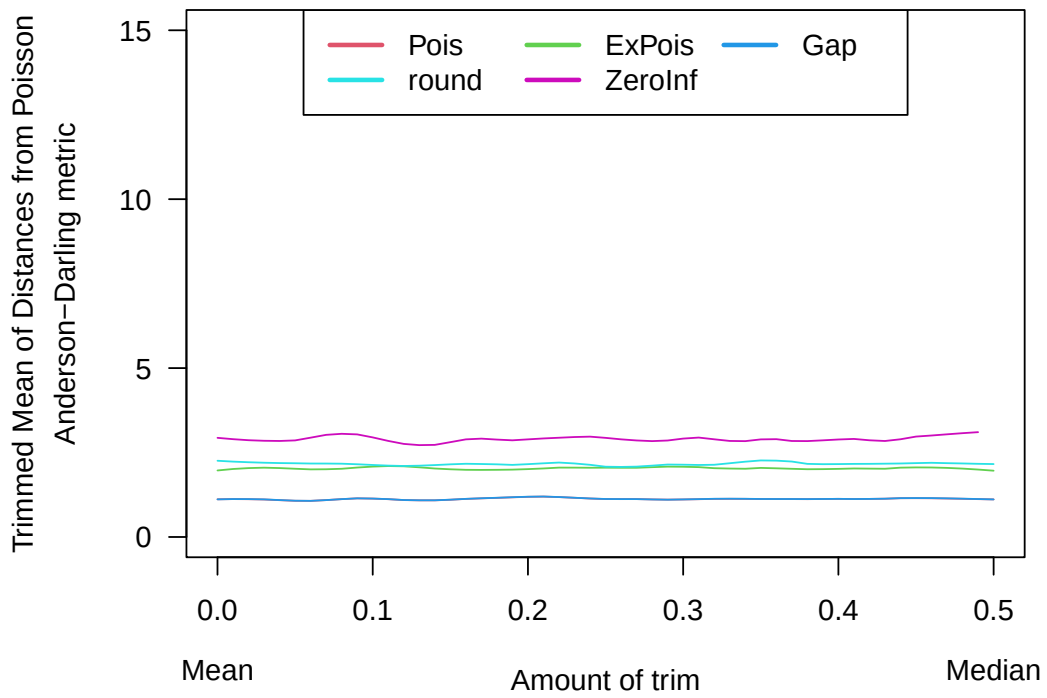


Figure 2: The mean for different trims for different distributions. The Anderson-Darling metric is used here. Curves have been smoothed using `loess` with a span of .2.